R4. Demand Forecasting for the New Product With No History

Real-world pain: Forecasting a product that has sold for two years is a solved textbook problem. The real money is in the new launch – the SKU with zero sales history, where a bad guess means either dead stock in a warehouse or a sold-out launch.

The problem: Forecast first-8-weeks demand for products that have **no sales history at all**, using only their attributes and the behaviour of similar past launches.

What we provide:

- 3 years of weekly sales for ~1,200 discontinued/mature SKUs, including each one's own launch curve
- Rich attributes: category, sub-category, colour, material, price band, brand tier, launch month, marketing spend flag
- 150 "new" SKUs with attributes only, and hidden actual first-8-week sales
- Store cluster definitions and a holiday calendar

Assumptions students may make:

- Analogous past products are a legitimate proxy – teams define the similarity logic
- Marketing spend, if flagged, is applied uniformly in the launch window
- Price stays constant during the first 8 weeks
- No competitor, weather, or social-media data exists

Desired output:

1. A weekly forecast for each new SKU, **with a prediction interval, not just a point number** – buyers need the range
2. `evaluate.py` score = weighted MAPE / WAPE on the hidden actuals, plus a calibration check on the intervals
3. A recommended initial buy quantity per SKU that explicitly trades off stock-out cost vs. markdown cost (teams state both costs)
4. An explanation of which attributes drove each forecast – a buyer must be able to argue with the number
5. A named list of the 10 SKUs the model is least confident about, and why

Stretch: allocate the buy quantity across store clusters rather than one national number.